B2B SALES ANALYTICS · LEAD FUNNEL

Paid and organic search bring in the deals.
Social brings in leads that rarely sign.

· Julian Batto-Hokson · 8,000 real MQLs · 842 signed sellers · Jun 2017 to Nov 2018

· Python · SQL · dbt · DuckDB · Streamlit · CrewAI · October 2026

Scope note: this is Olist's real seller-acquisition funnel. It records every marketing-qualified lead (MQL) and every lead that signed, but no deal amounts, pipeline stages or quotas. It can measure conversion and speed. It can't measure pipeline dollars, and because SDRs and closers are only recorded on won deals, it can't give a conversion rate per rep.

EXECUTIVE SUMMARY

Olist is a Brazilian marketplace that signs up merchants to sell through it. I modeled its published B2B funnel, 8,000 MQLs from June 2017 to May 2018 and the 842 that signed, into a dbt star schema on DuckDB, then built a Streamlit dashboard, a lead mix simulator and a local AI analyst on top.

Overall, 10.5% of MQLs signed, and the median deal closed 14 days after first contact. That headline hides a ramp: 2017 leads converted at 3.1% and waited a median of 216 days, while 2018 leads converted at 13.0% in 12.5 days. The sales team only started closing in December 2017.

Paid search (12.3%) and organic search (11.8%) convert above average and supply 48% of all MQLs. Social supplies 17% of MQLs but converts at 5.6%, and email at 3.0%. Their 95% intervals sit entirely below search's, so the gap isn't noise. In the simulator, moving 10 points of monthly volume from social to paid search adds about 10 deals a month at 2018 rates.

10.5%
MQL-to-deal conversion (842 of 8,000)

14 days
Median first contact to signed

13.0%
2018 lead conversion, vs 3.1% for 2017 leads

12.3%
Paid search conversion, best real channel

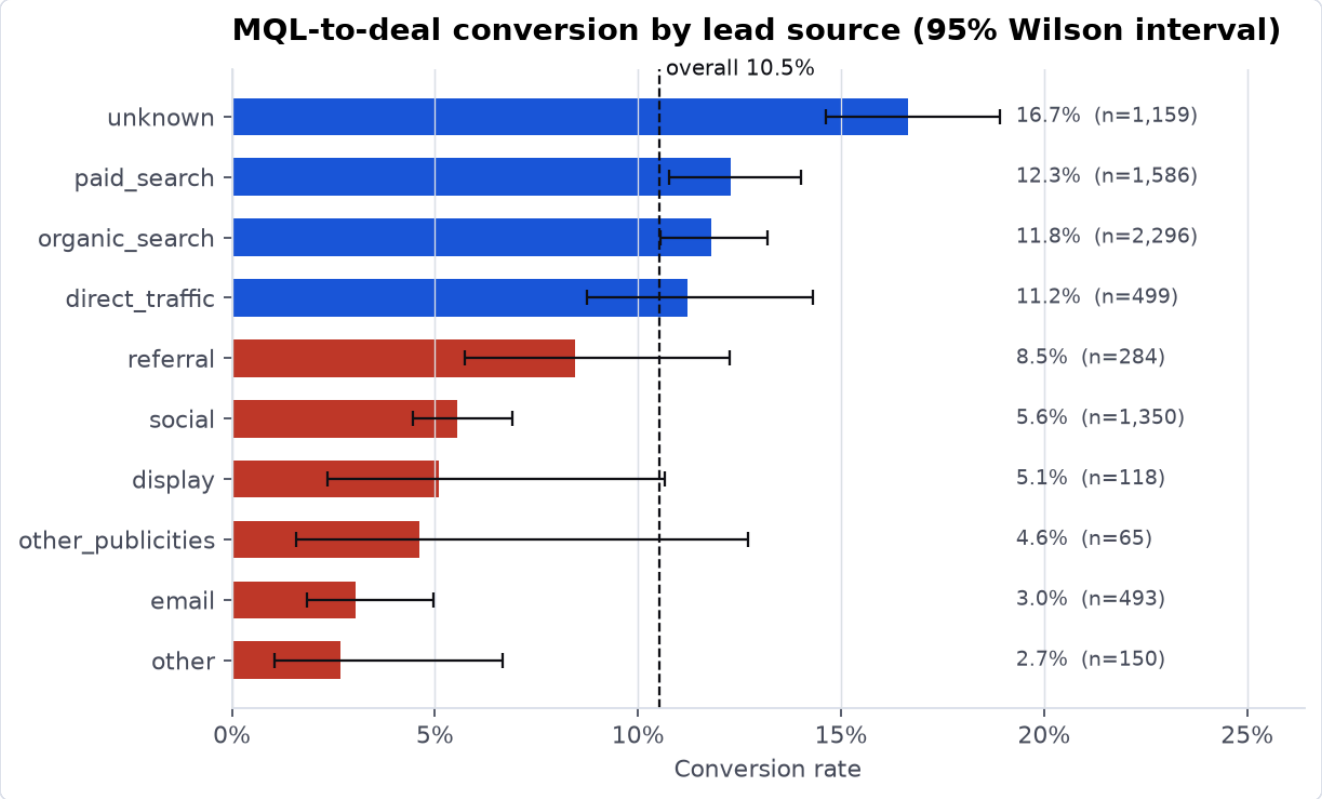
5.6%
Social conversion on 17% of MQL volume

17%
Deals sourced by the top SDR (1 of 32)

ANALYSIS

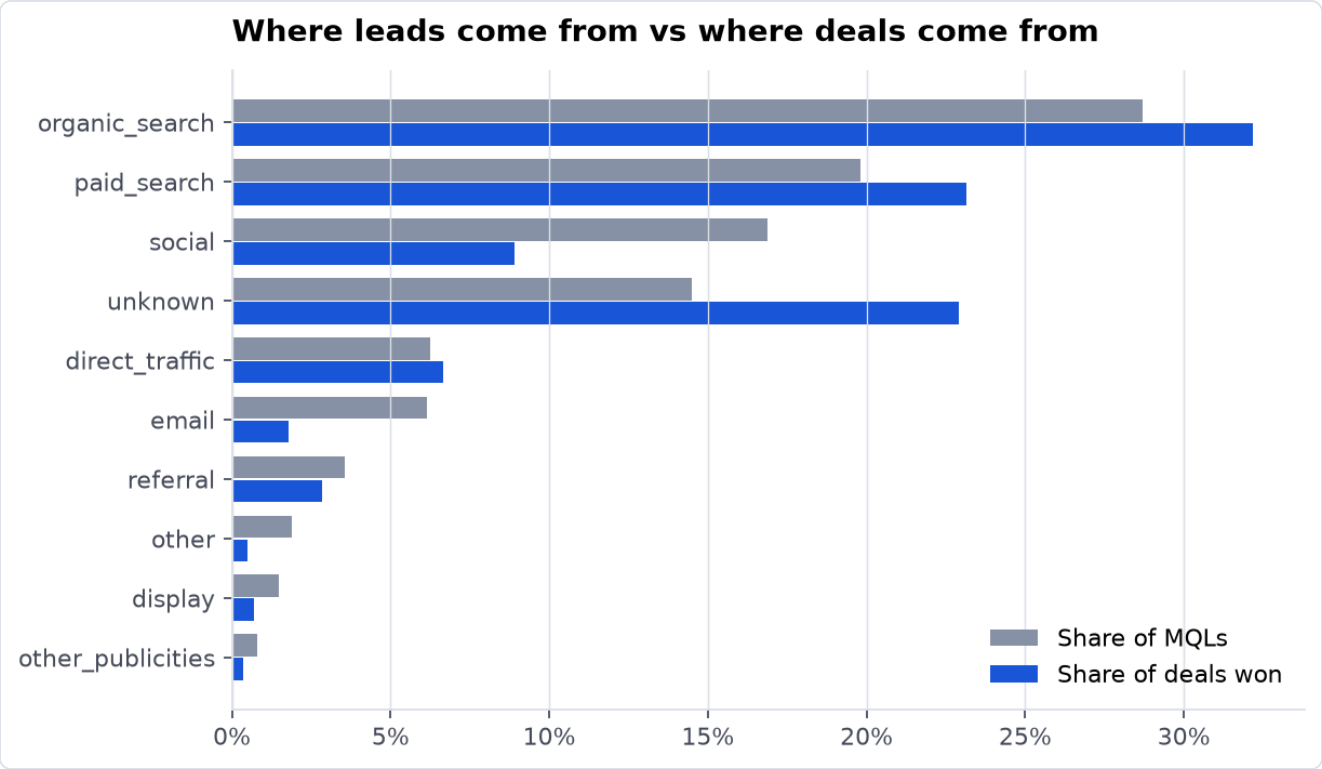
What the data shows

01 Search converts, social and email don't



Lead source	MQLs	Deals	Conversion	95% interval	Share of MQLs	Median days to close
paid_search	1,586	195	**12.3%**	10.8% to 14.0%	19.8%	15
organic_search	2,296	271	**11.8%**	10.6% to 13.2%	28.7%	14.5
direct_traffic	499	56	11.2%	8.7% to 14.3%	6.2%	10
referral	284	24	8.5%	5.7% to 12.3%	3.6%	18.5
social	1,350	75	**5.6%**	4.5% to 6.9%	16.9%	30
email	493	15	**3.0%**	1.9% to 5.0%	6.2%	21
unknown (no source recorded)	1,159	193	16.7%	14.6% to 18.9%	14.5%	11

Display, other_publicities and other (333 MQLs combined) are left out of the table: their intervals are too wide to say much. The full table is in `outputs/summary_by_lead_source.csv`.



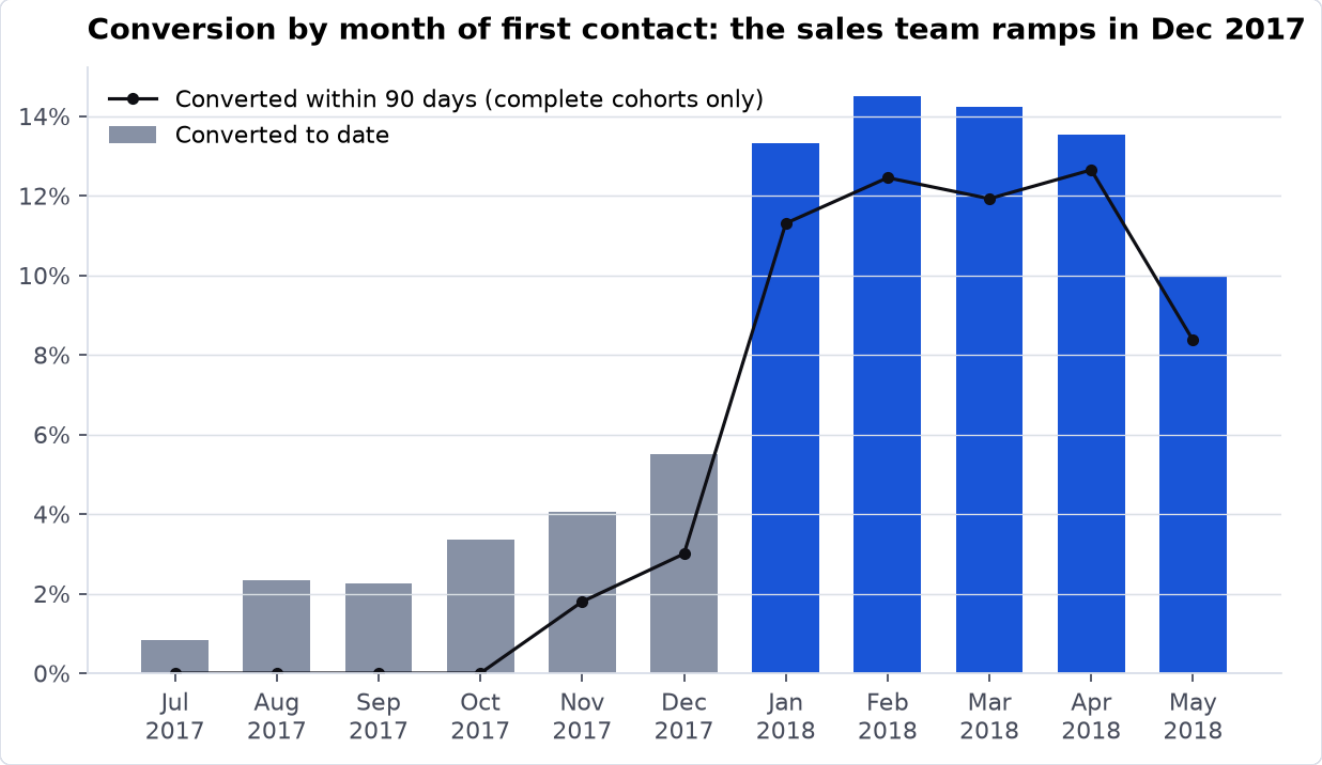
INTERPRETATION

Social generates 16.9% of leads but only 8.9% of deals. Organic search does the opposite: 28.7% of leads and 32.2% of deals. Social leads also take twice as long to sign when they do (median 30 days vs 14.5). The intervals are computed with the Wilson method in a dbt macro, so a small source can't look like a winner by luck.

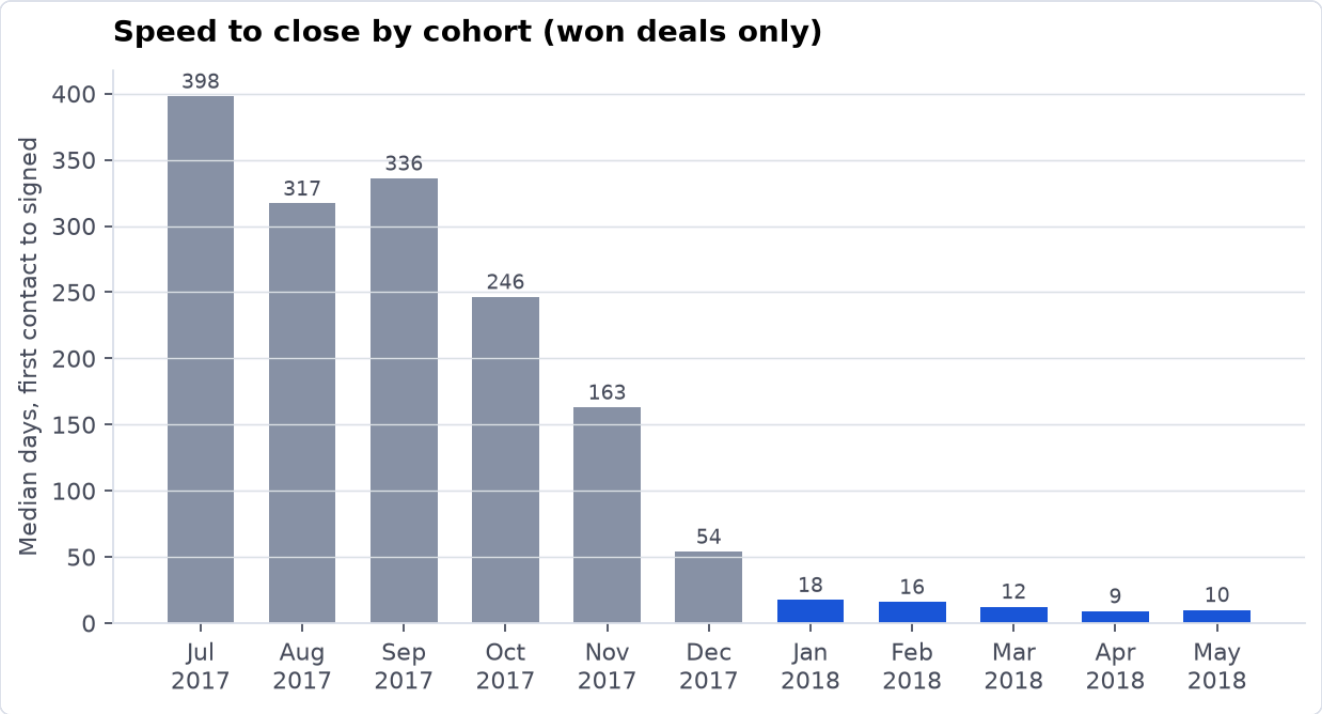
DATA-QUALITY FLAG, NOT A CHANNEL

The best-converting group is leads with no recorded source (16.7%). That could mean sales-sourced leads were never tagged. It's shown, but I don't recommend acting on it until someone finds out where those leads come from.

02 The funnel only started working in December 2017



Lead year	MQLs	Deals	Conversion	Median days to close
2017 (Jun to Dec)	2,002	61	3.1%	216
2018 (Jan to May)	5,998	781	13.0%	12.5



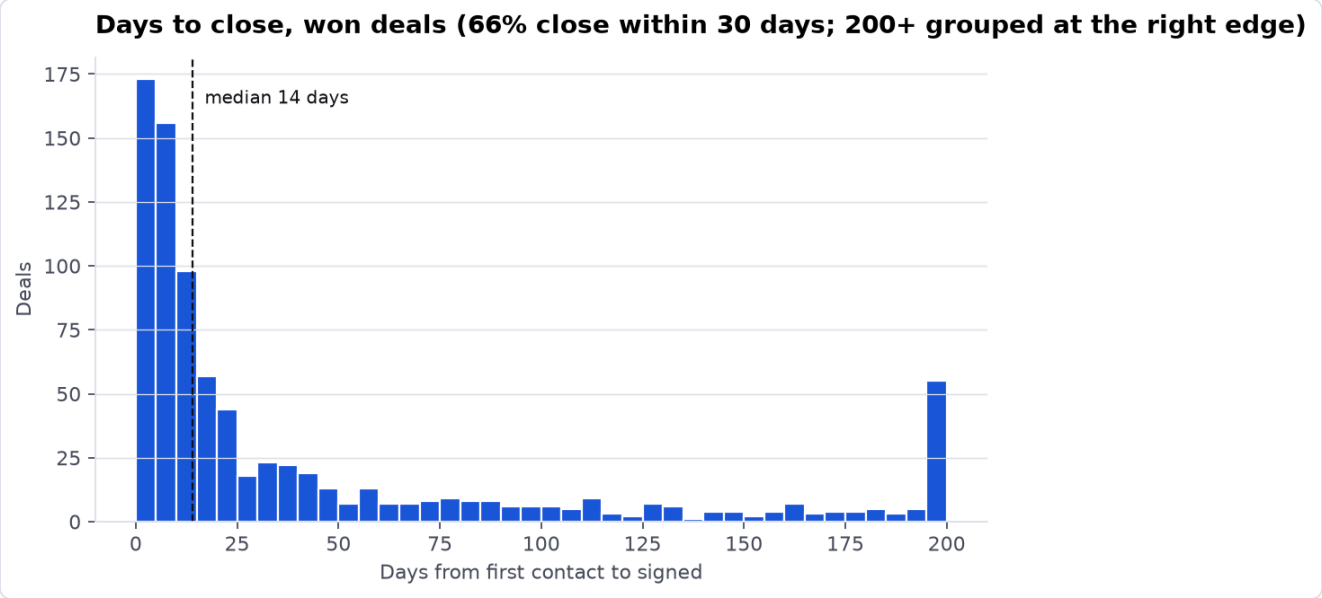
WHY THIS MATTERS FOR EVERY OTHER NUMBER

No deal closed before 2017-12-05. Leads from mid-2017 sat for months until a sales team existed to call them, so they convert worse and close slower for reasons unrelated to their source. Any channel or rep comparison that mixes 2017 and 2018 is distorted. The dashboard has a "2018 leads" filter and the simulator uses 2018 rates only.

CENSORING HANDLED

The 90-day line only appears for cohorts where every lead was at least 90 days old at the as-of date (2018-11-14). The May 2018 cohort's 180-day rate is left blank for the same reason. Without these window flags, young cohorts look worse than they are.

03 Two-thirds of won deals sign within a month

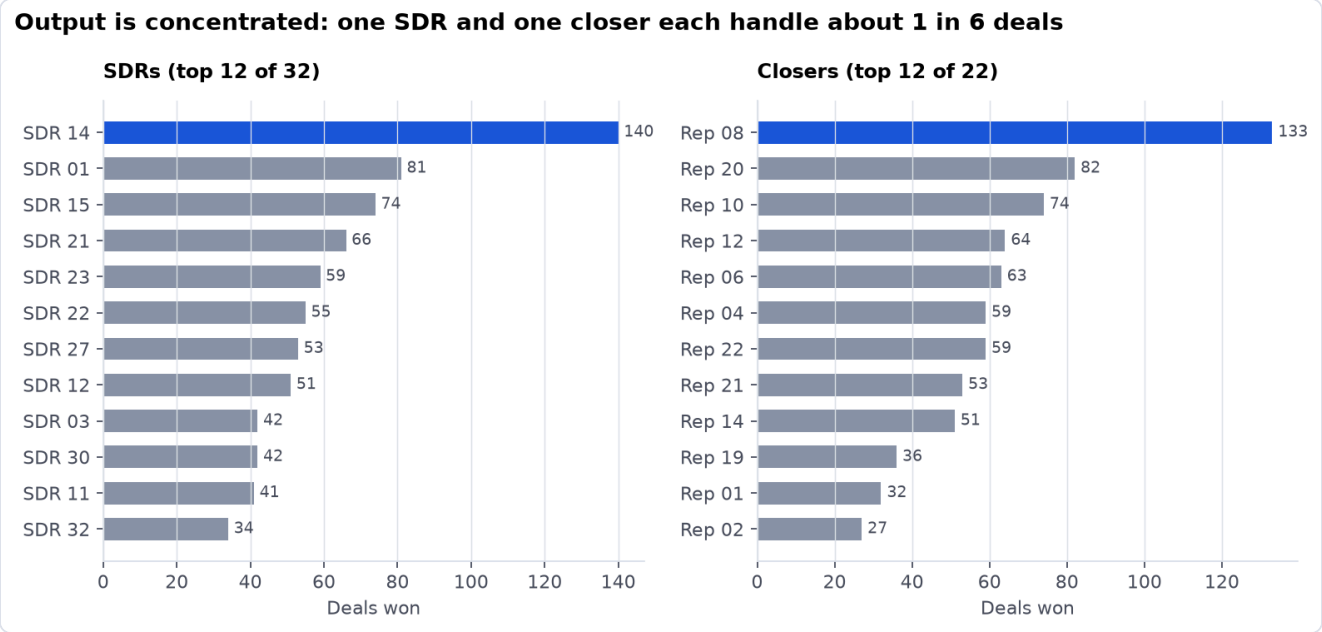


554 of 842 won deals (66%) signed within 30 days of first contact. The median is 14 days, but the mean is 48.5 because of the 2017 backlog. One deal is dated 2 days before its first contact. It's flagged and excluded from cycle metrics, and a dbt test enforces that.

WHAT THIS SUGGESTS

Among 2018 leads, 551 of 781 wins (71%) came within 30 days, and 144 (18%) came after day 60. Most of the value is in fast follow-up. Whether late chasing is worth SDR time is a question to test, since 18% of wins is not nothing.

04 Rep output is concentrated in one SDR and one closer



SDR 14 sourced 140 deals (16.6%) and Rep 08 closed 133 (15.8%), across 32 SDRs and 22 closers. IDs in the source data are anonymized hashes; the readable labels are assigned in dbt.

WHAT THIS CAN AND CAN'T SAY

These are volume and speed, not performance. Reps are only recorded on won deals, so a rep with many wins might also have many losses. A real CRM with rep ownership on every lead would allow per-rep conversion.

05 Home decor and health and beauty sellers sign most often



Home decor (105 deals, median 13 days) and health and beauty (93, median 12) lead. Audio and video electronics sellers are slower (64 deals, median 22 days). Only 45 of 842 deals have a declared monthly revenue, so seller size isn't analyzed.

PLANNING TOOL

Lead mix simulator

Expected deals = planned MQLs from each source × that source's 2018 conversion rate. The low and high bounds use each source's 95% interval. With no levers moved at the historical volume, it reproduces the 842 actual deals exactly (a test checks this).

Scenario (1,200 MQLs a month, 2018 average)	Expected deals / month	Planning range	Blended conversion
Baseline: 2018 mix	156.3	132.5 to 185.9	13.0%
Move 10 points of volume from social to paid search	**166.5**	142.0 to 196.8	13.9%

Scenario (1,200 MQLs a month, 2018 average)	Expected deals / month	Planning range	Blended conversion
Lift every source's conversion by 10% (faster follow-up)	**171.9**	145.8 to 204.5	14.3%

READ IT AS A PLANNING BAND

The range assumes every source lands at its bound at once, so it's wide on purpose. It also assumes extra paid search spend buys leads that convert like today's, which gets less true as spend scales.

LOCAL AI

A free AI analyst that has to show its SQL

The dashboard's copilot is a CrewAI agent running on a local Ollama model (llama3.2, 3B), so it has no API cost. It answers by writing SQL against the marts. Every query and result is shown next to the answer, and a fact checker flags any number in the prose that didn't come from a query result.

Question (live test, llama3.2)	SQL	Answer	Caught by guardrails?
Which lead source converts best?	Correct	**Right**: unknown, 0.17	n/a
What share of MQLs became deals?	Ran, wrong column	**Wrong**: called each source's share of leads its "conversion rate"	**No**: the numbers exist in the result, so the fact check passed them
Which SDR closed the most deals, and how fast?	12 failed queries	**Wrong**: right SDR, invented counts (120 deals vs 140)	**Yes**: marked ungrounded after a retry, 3 invented figures flagged
Median days to close by segment?	Correct	**Right** values, but listed most of them without segment names	n/a

WHAT THE LIVE TEST SHOWS

A free 3B model gets simple questions right and fails in two different ways on the rest. When its SQL fails, the guardrails catch it: the answer is marked ungrounded and every invented number is listed. When its SQL works but it misreads a column, nothing catches it, because every number it quotes is real. That second failure is why the app always shows the query and result table next to the prose. The unedited log is in `docs/live_eval_llama3.2_2026-10-20.txt`.

Guardrail	How it's enforced
Read-only	DuckDB's parser must see exactly one SELECT; the connection is read-only and blocks reading local files
Bounded	200-row cap and a 15-second timeout per query
Grounded	Answers with no successful query are retried once, then flagged
Fact-checked	Numbers in the answer that match no query result are listed under the answer

NEXT STEPS

What I'd recommend, and what I'd test first

CHANNEL MIX

Shift budget from social toward paid search, in steps

Social converts at half the average. Move spend in 5-point steps and check that new paid search leads still convert near 14.8% (their 2018 rate).

```
Simulator: +10 points to paid search ≈ +10 deals / month
```

DATA

Find out what "unknown" source leads are

1,159 leads with no source convert best of all. If they're referrals or sales-sourced, that channel deserves investment, but it's invisible today.

```
Audit lead capture for the 1,099 tagged "unknown" and 60 blank
```

PROCESS

Test a 60-day follow-up cutoff

71% of 2018 wins closed inside 30 days, 18% after day 60. Split leads older than 60 days into "keep chasing" and "recycle" groups and compare.

```
Track: wins after day 60 per SDR hour spent
```

MEASUREMENT

Record the SDR on every lead, not just wins

Without it, rep coaching relies on volume. With it, per-rep conversion and speed become measurable.

```
One field added at lead assignment
```

TECHNICAL REFERENCE

Data and pipeline

Dimension	Value
Source	Marketing Funnel by Olist (Kaggle, CC BY-NC-SA 4.0), real and anonymized
Rows	8,000 MQLs (first contact 2017-06-14 to 2018-05-31); 842 closed deals (2017-12-05 to 2018-11-14)
Not included	Deal amounts, pipeline stages, activity logs, quotas, reps on lost leads
Pipeline	Hash-verified fetch → typed DuckDB load with integrity checks (rolls back on any failure) → dbt staging and marts
Marts	fct_lead_funnel, agg_lead_source_conversion, agg_cohort_conversion, dim_sdrs, dim_sales_reps, agg_segment_summary, agg_landing_page_conversion, dim_date
Testing	62 dbt checks including 7 reconciliation tests; 125 pytest tests; GitHub Actions CI; full rebuild run 3 times with identical table hashes
Tools	Python (pandas), SQL, dbt-duckdb, DuckDB, Streamlit, Plotly, Matplotlib, CrewAI, Ollama, pytest